

OMUSOLOSMART

Taxpayer Conversational AI for the URA Service Environment

Strategic Engagement Briefing, CX Insights, Expectations, System Readiness Audit, and Demonstration Protocol

Presenting Entity: School of Computing and Informatics Technology (SCIT), Makerere University & Sunbird AI

Host Audience: Uganda Revenue Authority (URA) — Client Experience (CX) & Virtual Support Channels

Date of Engagement: September 2026 • Fiscal Year 2026/27 • Document Version 2.5 (Consolidated Full-Context Edition)

Official Engagement Mandate: URA Client Experience Internal Memo

“As part of our ongoing efforts to enhance Client Experience through innovative and technology-enabled solutions, we will be hosting a team of software engineers from the School of Computing and Informatics Technology, Makerere University, to present OmusoloSmart, a conversational AI assistant developed specifically for the URA taxpayer service environment.

OmusoloSmart enables taxpayers to ask tax-related questions through voice or text in Luganda, Swahili, and English, and receive instant responses based strictly on verified information from official URA documentation, 24 hours a day.

This aspect, in particular, may be of interest to our Client Experience function given our virtual support channels.

You are therefore invited to the engagement to explore the solution, understand its capabilities and potential application within our support channels, and provide your insights from a Client Experience perspective. So, come prepared to actively engage with the presenters, ask questions, and provide your feedback and recommendations.”

1 Executive Summary & Strategic Positioning

OmusoloSmart is Uganda’s first sovereign, multilingual conversational intelligence platform engineered exclusively for public revenue administration. Developed by researchers and software engineers at Makerere University’s School of Computing and Informatics Technology (SCIT) in collaboration with Sunbird AI, the system represents an architectural paradigm shift from generic commercial large language models (LLMs) to a **statutory-grounded, epistemically disciplined revenue assistant**.

1.1 Strategic Positioning Pillars

- **Not an Off-the-Shelf Commercial Wrapper:** Unlike generic consumer chatbots (e.g., ChatGPT, Claude) that hallucinate tax figures or produce unverified legal advice, OmusoloSmart enforces **mathematical and statutory grounding**. Inquiries involving rates, thresholds, and calculations route through deterministic tax engines grounded in versioned FY2026/27 Acts and Tariffs.
- **Strict Epistemic Discipline:** The system operates at greedy zero-temperature with real-time numeric contradiction withholding: if a drafted answer conflicts with official documentation, it is withheld before reaching the taxpayer.
- **Tricameral Linguistic Delivery:** Delivers complete functional parity in **English, Luganda, and Swahili** across both voice and text interfaces, breaking down linguistic and digital literacy barriers for Uganda’s informal sector.

2 Stakeholder Landscape & Audience Profile

Under the government’s Domestic Revenue Mobilization Strategy (DRMS), the URA is actively pivoting from an enforcement-centric agency toward a **service-oriented, taxpayer-centric facilitator**. The URA Client Experience department operates virtual support channels, including the Contact Centre (toll-free 0800 117 000 / 0800 217 000), WhatsApp (0772 140 000), web portal live chat, and walk-in service centers across Uganda.

2.1 Audience Profile & Stakeholder Interests

- **Contact Centre Operations Managers:** Focused on First Contact Resolution (FCR), Average Handling Time (AHT), queue wait times, and preventing agent burnout during filing deadlines.
- **Taxpayer Education & Formalization Leads:** Focused on procedural clarity, step-by-step onboarding, and bringing informal traders (Kikuubo wholesalers, boda-boda operators, market vendors) into voluntary compliance.
- **Digital Support & Omni-Channel Specialists:** Dedicated to ensuring feature and accuracy parity across the Web Portal, WhatsApp, Mobile App, and telephony IVR.
- **Compliance, Legal & Audit Observers:** Concerned with statutory reliability, strict citation of tax legislation, prevention of unauthorized tax advice, and adherence to national data sovereignty.

Operational Challenge	Current CX Pain Point at URA	OmusoloSmart Resolution
Filing Deadline Spikes	On the 15th of each month (PAYE/VAT) and Dec 31 (Income Tax), inbound queues surge by over 400%, leading to dropped calls and high AHT.	Automates Level-1 repetitive inquiries with sub-second response times ($p_{50} < 700\text{ms}$ on cache/rates), maintaining 100% availability.
Linguistic Divide	The informal sector (over 50% of Uganda’s GDP) conducts business in Luganda and Swahili, but statutory forms are in dense bureaucratic English.	Delivers trilingual voice and text interaction with native Luganda and Swahili fine-tuning, translating complex tax concepts into local idioms.
Statutory Liability Risk	Generic AI models hallucinate outdated thresholds (e.g., citing superseded UGX 235,000 PAYE threshold instead of FY2026/27 UGX 335,000).	Uses versioned deterministic tax rate tables and contradiction withholding: wrong figures are blocked before reaching the user.
Frustrating Bot Loops	Traditional rule-based IVR trees and rudimentary keyword bots trap distressed taxpayers in circular dead-ends without recourse.	Employs automated distress detection (anxiety, hardship, penalty fears) that triggers compassionate de-escalation and warm officer handoffs.

Figure 1: URA CX Operational Bottlenecks vs. OmusoloSmart Technical Interventions

3 The Four Architectural Pillars & Structural Innovations

1. **Epistemic Safety & Zero-Drift Grounding:** Statutory inquiries (rates, thresholds, and calculations) bypass generative token guessing entirely. Queries are routed through deterministic calculators executing versioned FY2026/27 URA rate tables (`tables.py`). A real-time claim verification engine compares LLM assertions against cited passages; if an assertion contradicts the legal passage (e.g., claiming VAT is 20% when the law states 18%), the system actively intercepts the reply, suppresses the wrong figure, and serves the official URA text.
2. **Linguistic Parity & Native Vernacular Voice:** Developed in collaboration with Sunbird AI, the speech stack utilizes **Whisper-Large-v3-SALT** for Automatic Speech Recognition (ASR) and **Spark-TTS-SALT** for Text-to-Speech (TTS). Speech inference operates on CUDA at a **0.129x Real-Time Factor** (processing audio 7.7 times faster than real-time). Vernacular numeral preservation ensures that East African monetary idioms (*“obukadde 150”* , *“milioni 37.5”* , *“emitwalo ebiri”*) are parsed accurately without collapsing to English, while disambiguating the Luganda action verb *“Saba”* (*to apply/request*) from the Swahili numeral *“saba”* (*7*).
3. **Empathetic Warm Escalation (Anti-Frustration Mechanism):** Conversations are continuously evaluated for taxpayer distress (hardship, business closure, frozen accounts, penalty anxiety). Rather than abandoning the taxpayer in repetitive loops, OmusoloSmart compiles a structured **Ticket Context Packet** (full transcript, legal context, sentiment analysis, and opening advice) for seamless takeover by a human Contact Centre agent, reducing officer triage from 8 minutes to under 30 seconds.
4. **Sovereign On-Premises / National Cloud Deployment:** OmusoloSmart complies strictly with the **Uganda Data Protection and Privacy Act 2019 (PDPO)**. The entire stack runs on local infrastructure (Docker / Kubernetes on sovereign cloud, e.g., Crane Cloud / NITA-U). No taxpayer queries or voice audio files are transmitted to foreign commercial APIs. All interactions are audited through an append-only cryptographic Merkle ledger.

3.1 Supplementary Capabilities: Offline Edge & Omni-Channel Adaptability

- **Standalone Offline Capability:** Includes an exportable, lightweight offline engine (FAISS vector store + SQLite) capable of running on low-power local workstations at remote border posts and regional offices without internet access.
- **Headless Omni-Channel Delivery:** Standardized REST, W3C Server-Sent Events (SSE), and duplex WebSockets (`/v1/chat`, `/v1/chat/stream`, `/v1/voice/chat/stream`) allow the exact same core intelligence to power Web Chat (Next.js), WhatsApp Business API, and Contact Centre telephony switches (SIP / Asterisk / FreeSWITCH).

4 Expectations from Both Sides

4.1 What the URA Client Experience Team Expects

- **A Functional System, Not Slideware:** Live execution of real taxpayer queries submitted in natural, messy, conversational language.
- **Authentic Ugandan Vernacular:** Verification that the AI understands authentic Luganda and Swahili, local commercial slang (*“kikuubo”*, *“(omusalaba)”*), and bilingual code-switching (*“(Luglish)”*).
- **Tangible Queue Reduction:** Proof that the system can meaningfully deflect routine inquiries and lower queue times without increasing escalations.
- **Integration Simplicity:** Clear architectural pathways to embed into `ura.go.ug`, the URA Mobile App, WhatsApp, and telephony channels.
- **Risk & Governance Transparency:** Explicit demonstrations of how the system refuses out-of-scope inquiries, handles tax evasion prompts, and protects citizen privacy.

4.2 What Makerere SCIT Must Prepare to Deliver

- **Address Accuracy Skepticism:** Tax officers will probe edge cases (e.g., input VAT deductibility on passenger vehicles, withholding tax exemptions, bad debt relief). Presenters must highlight the deterministic rate tables and cited statutory provisions.
- **Alleviate Job Security Concerns:** Contact Centre staff may fear automation. Frame OmusoloSmart explicitly as an **officer co-pilot and force multiplier** that eliminates routine drudgery and prepares structured case briefs.
- **Demonstrate Voice & Acoustics Live:** Presenters must use live microphone input and external speakers to showcase real-time Luganda and Swahili voice recognition and synthesis.
- **Expose Back-Office Diagnostic Views:** Present the supervisor ticket queue, sentiment analytics, and cryptographic audit ledger.

5 System Readiness & Empirical Audit Results

OmusoloSmart has completed extensive full-stack benchmarks executed over the live public gateway (ngrok) on dedicated sovereign hardware (NVIDIA RTX A6000 48GB GPU, GPU #4). All reported metrics are verified against checkable report artifacts in the codebase.

5.1 Flagship 2,000-FAQ Benchmark & Empirical Comparison

5.2 Recent Engineering Enhancements Driving Top-Tier Accuracy (99.85%)

- **Dynamic Damerau-Levenshtein Typo Correction:** Implemented length-adaptive fuzzy distance matching (≤ 2 edit distance for terms ≥ 7 characters, ≤ 1 for terms 4..6 characters, and double-letter acronym squeezing: `tinn` $\rightarrow$ `tin`, `vatt` $\rightarrow$ `vat`, `pinn` $\rightarrow$ `pin`). Common taxpayer misspellings (`calcultate`, `registratin`, `environmntal`, `witholdng`, `customos`, `incom`, `retun`, `objectionn`) are resolved before semantic search without altering user intent.

Evaluation Metric	2,000 FAQs (Flagship)	1,000 FAQs	300 FAQs	Audit Status & Target
Overall Factual Accuracy	99.85% (1,997/2,000)	100.0% (1,000/1,000)	100.0% (300/300)	PASSED (Target > 99%)
English (en) Accuracy	100.0% (1,529 turns)	100.0% (480 turns)	100.0% (100 turns)	PASSED (Target > 99%)
Luganda (lg) Accuracy	99.58% (238/239)	100.0% (264 turns)	100.0% (100 turns)	PASSED (Target > 95%)
Swahili (sw) Accuracy	99.14% (230/232)	100.0% (256 turns)	100.0% (100 turns)	PASSED (Target > 95%)
Domestic Taxes Accuracy	100.0% (550 queries)	100.0% (423 queries)	100.0%	PASSED (Target > 95%)
Customs & Trade Accuracy	100.0% (450 queries)	100.0% (252 queries)	100.0%	PASSED (Target > 95%)
Transport & Motor Vehicles	100.0% (350 queries)	—	—	PASSED (Target > 95%)
EFRIS & E-Invoicing	99.43% (348 queries)	—	—	PASSED (Target > 95%)
Tax Education Accuracy	99.67% (299 queries)	100.0% (325 queries)	100.0%	PASSED (Target > 95%)
Conversational Grade	99.10%	98.09%	99.93%	PASSED (Target > 95%)
Emotional Intelligence (EQ)	99.25%	95.56%	99.76%	PASSED (Target > 95%)
Official Contact Integrity	0 False Redactions	0 False Redactions	0 False Redactions	PASSED (Zero over-redaction)
Speech STT (Whisper-SALT)	Verified (0.129x RTF)	3/3 Verified	3/3 Verified	PASSED (Real-time speed)
Speech TTS (Spark-TTS)	Verified (Trilingual)	3/3 Verified	3/3 Verified	PASSED (Natural intonation)
HTTP Availability (200 OK)	100.0% (2,000/2,000)	100.0%	100.0%	PASSED (0 server errors)
Median Latency (p_{50})	388.2 ms (0.39s)	3.72s	7.53s	EXCELLENT (< 500 ms)
Sustained Throughput	2.32 QPS	1.02 QPS	—	PASSED (Burst: 29.4 req/s)

Table 1: Consolidated Empirical Results across 3,400 Total Audited Queries over Public Ngrok Gateway

- Universal Currency Parsing & Standalone Amount Execution:** Enhanced regular expression engines parse unformatted currency amounts across all regional notations (UGX 2000000, 2000000 ugshs, 2000000shs, 2000000/=, 2m ugx) without requiring comma delimiters. Standalone currency inputs immediately trigger verified multi-tax computations across PAYE, VAT (18%), and WHT (6%) under the in-force FY2026/27 rate tables.
- Citizen Procedural Routing (PRN, TCC, PWD Relief):** Ingested verified operational workflows for Payment Registration Number (PRN) generation and 21-day reactivation via mobile money (*165# MTN / *185# Airtel), Tax Clearance Certificate (TCC) electronic issuance, and Section 21(1)(v) Income Tax Act statutory exemptions for Persons with Disabilities (first UGX 1,460,000/month exempt).
- Low-Bandwidth Adaptations & Emergency Offline Portal:** Engineered client-side reactive network status monitoring (W3C NetworkInformation API), GZip compression (> 500 bytes threshold), and an offline portal fallback (offline.html) with client-side tax calculators to ensure continuous access in rural areas and during network outages.
- Sentence Splitting & Contradiction Withholding Repair:** Corrected PDF OCR spacing artifacts on decimal figures (e.g. 37. 5 million → 37.5 million) and refined NLI claim verification to evaluate assertions against the full retrieved context, eliminating false contradiction withholdings.

5.3 Live Multilingual Concurrency Audit: Load, Volume, Stress, and Spike Profiles

To establish operational robustness, reliability, and resilience under concurrent demand, live multi-client testing was executed over the public gateway (ngrok) with traffic distributed across English (en), Luganda (lg), and Kiswahili (sw). The audit

evaluated four distinct execution profiles covering sustained nominal traffic, heavy multi-clause enterprise inquiries, stepped saturation ramping, and unannounced shock bursts.

Concurrency Profile	Concurrency	Requests	Success Rate	Throughput	p_{50} / p_{95} (ms)	Language Parity (en / lg / sw)
Load Test (Sustained)	6 VUs	60	100.0% (60/60)	1.97 QPS	342.0 / 14,180	100% / 100% / 100% (p_{50} : 326 / 1,525 / 374 ms)
Volume Test (Heavy)	8 VUs	45	100.0% (45/45)	2.82 QPS	411.6 / 13,153	100% / 100% / 100% (p_{50} : 408 / 414 / 396 ms)
Stress Test (Ramped)	4→24 VUs	52	100.0% (52/52)	10.21 QPS	680.0 / 925.6	100% / 100% / 100% (p_{50} : 720 / 720 / 545 ms)
Spike Test (Shock Burst)	32 VUs	32	100.0% (32/32)	6.29 QPS	727.4 / 5,046	100% / 100% / 100% (p_{50} : 725 / 727 / 755 ms)
Total Audited Profile	Up to 32 VUs	189	100.0% (189/189)	10.21 Peak	342.0 (p_{50} load)	Zero 5xx errors; 100% statutory figure retention

Table 2: Empirical Multilingual Concurrency, Volume, Stress, and Spike Benchmark Results

Key Reliability and Resilience Invariants Verified:

- **Zero Server Failures (100% Availability):** Across all 189 concurrent requests spanning 4 stress profiles, the platform returned zero HTTP 5xx errors, zero connection drops, and zero aborted streams through the public tunnel.
- **Peak Elastic Throughput (10.21 QPS):** During stepped concurrency scaling (4 → 24 VUs), the backend reached 10.21 queries per second with a 680 ms median response time (p_{95} = 925.6 ms), demonstrating stable asynchronous worker scheduling.
- **Instantaneous Shock Burst Resilience:** When subjected to an unannounced 32-request concurrent burst (Spike Profile), the system absorbed the shock with a 727.4 ms median latency and immediate post-burst recovery, maintaining 100% accuracy without queue starvation.
- **Trilingual Semantic & Figure Parity:** All three national languages sustained 100% factual resolution (EN: 60/60, LG: 73/73, SW: 56/56), with statutory rate figures (e.g., 18% VAT, 6% WHT, progressive PAYE bands) fully preserved across vernacular Machine Translation boundaries.

6 Client Experience Value Propositions & Impact

CX Metric	Operational Impact Achieved by OmusoloSmart
First Contact Resolution (FCR)	Routine level-1 inquiries (TIN registration, return due dates, tax rates, PRN generation, clearing steps) are resolved on the first interaction without officer triage.
Queue Deflection	Deflects an estimated 65–75% of routine inbound volume from telephone lines and live chat queues, eliminating taxpayer queue abandonment.
Average Handling Time (AHT)	For escalated complex turns, the automated Ticket Packet (carrying transcript, legal context, sentiment analysis, and opening advice) reduces human officer research time from ~8 minutes to under 45 seconds.
24/7/365 Service Continuity	Extends full taxpayer support outside standard civil service hours (nights, weekends, public holidays), allowing taxpayers to prepare returns at their convenience.
Informal Sector Formalization	Lowens the friction of tax education by explaining tax obligations in conversational Luganda and Swahili, empowering informal traders to voluntarily register and comply.

Figure 2: Direct Operational Benefits for URA CX Leadership

7 Structured Live Demonstration Protocol

To ensure maximum credibility during the presentation, the Makerere team should execute a tightly rehearsed, 4-phase demonstration showing **live execution**:

Phase 1: The Modern Digital Taxpayer — English Web Chat

Objective: Demonstrate step-by-step procedural clarity, UI bold anchors, and due date callouts.

Action: In the web chat interface, submit the following query in English:

"How do I file my annual income tax return and when is it due?"

Expected Live Output:

To file your annual tax return:

1. Login ura.go.ug with TIN/password → e-services → e-returns → select return type and download template
2. Enable macros, fill without renaming or copy/paste
3. Validate to generate upload file
4. Back to e-returns upload file with return period and captcha
5. Submit
6. E-acknowledgment is issued (also emailed/portal)

Due date: Final returns are due within 6 months after the end of the financial year (31st December for standard fiscal year).

If you get stuck at any step, URA is happy to help: visit <https://ura.go.ug>, call toll-free 0800 117 000 / 0800 217 000...

Phase 2: The Informal Market Trader — Native Luganda Voice

Objective: Demonstrate Whisper-SALT speech recognition, Luganda procedural generation, and Spark-TTS voice.

Action: Click the voice input microphone and speak naturally in Luganda:

"Nkola ntya okwewandiisa okufuna TIN y'omuntu kinnoomu ku mutimbagano?"

Expected Live Output:

Okwewandiisa okufuna TIN ey'Amangu ng'omuntu kinnoomu:

1. Genda ku ura.go.ug
2. Koonna ku **Funa TIN**
3. Londa **TIN** ey'Amangu
4. Londa **Omuntu kinnoomu**
5. Teekamu NIN yo n'obubaka bwo obukukwatako
6. Kakasa nti siyoboti
7. Waayo

Bw'oba ng'osanze obuzibu bwonna mu mitendera gyonna, URA yeesunga okuyamba: genda ku <https://ura.go.ug>...

→ **Audio Output:** The system synthesizes and plays back natural Luganda speech via Spark-TTS-SALT.

Phase 3: Cross-Border Commercial Trader — Swahili Statutory Calculation

Objective: Demonstrate cross-lingual statutory calculation and legal reference preservation.

Action: Submit a complex VAT and customs inquiry in Swahili:

"Kiwango cha kodi ya ongezeko la thamani (VAT) nchini Uganda ni asilimia ngapi na kiasi gani cha mauzo kinahitaji usajili?"

Expected Live Output:

Kiwango cha kawaida cha Kodi ya Ongezeko la Thamani (VAT) nchini Uganda ni **18%** kulingana na Sheria ya Kodi ya Ongezeko la Thamani (VAT Act).

Kiwango cha chini cha mauzo kinacholazimu usajili wa lazima wa VAT ni **Shs. 150,000,000** kwa mwaka (au Shs. 37.5 milioni katika miezi mitatu mfululizo).

Ikiwa utakabiliwa na changamoto yoyote katika hatua yoyote, URA iko tayari kukusaidia: tembelea <https://ura.go.ug>...

Phase 4: Distressed Taxpayer Seeking Relief — Empathetic Human Escalation

Objective: Demonstrate emotional intelligence (EQ), hardship sentiment detection, and automated ticket triage.

Action: Submit a distressed prompt simulating financial vulnerability:

"My bank account was frozen by URA and I cannot afford food for my children. What can I do?"

Expected Live Output:

Sentiment Identified: Hardship • **Escalation State:** Active

"I'm sorry you're going through this — let's look at what options you have."

Provides statutory options under the Tax Procedures Code Act (payment plan applications, formal objection within 45 days, and applying to the Commissioner General for release of agency notices).

→ **CX Action:** Displays official Contact Centre channels and generates a backend ticket containing the full transcript and context for an urgent officer callback.

8 Strategic Positioning Matrix: OmusoloSmart vs. Generic AI

Evaluation Dimension	Generic LLM / Commercial Wrapper	OmusoloSmart (Makerere SCIT / Sunbird)
Statutory Grounding	Hallucinates outdated figures; trained on open web blogs and superseded legal texts.	Grounded strictly in versioned URA FY2026/27 tables, Acts, and authoritative handbooks.
Contradiction Guard	Confidently serves conflicting rates or dates if asked with persuasive leading prompts.	Active numeric contradiction withholding: halts and suppresses unverified figures.
Ugandan Languages	Broken Luganda; invents non-existent words or mixes Coastal Swahili syntax.	Native Luganda and Swahili fine-tuned on local SALT corpora by Ugandan linguists.
Voice Processing	Robotic generic foreign English; cannot transcribe or synthesize indigenous languages.	Native Whisper-SALT ASR and Spark-TTS-SALT voice models supporting local accents.
Data Sovereignty	Transmits sensitive citizen inquiries to overseas servers (US/EU), violating PDPO.	100% on-premises or sovereign cloud hosting (Crane Cloud / NITA-U); zero foreign data egress.
Contact Centre Triage	Stands alone; cannot create tickets, track SLAs, or hand off context to officers.	Seamlessly integrated with ticket queues, providing structured context packets to agents.

Table 3: Architectural & Governance Comparison: Generic Solutions vs. OmusoloSmart

9 Anticipated Questions & Defensible Responses

Presenters should prepare for probing questions across all four operational, legal, linguistic, and infrastructural disciplines, synthesizing the full context of both analytical engagement reviews:

9.1 Category A: Client Experience (CX) & Contact Centre Operations

Q1: How does OmusoloSmart handle an inquiry during peak return periods when thousands of taxpayers hit the portal simultaneously?

Defensible Response:

“The architecture is decoupled into high-throughput microservices. Routine rate lookups, procedure requests, and FAQ inquiries are served via an in-memory Redis semantic cache and deterministic calculator router in sub-second response times ($p_{50} < 700\text{ms}$). Generative RAG turns run with batched inference and concurrency governors to ensure server stability. Under burst testing at 30 concurrent users, the system maintained 100% HTTP availability with zero dropped requests.”

Q2: How does OmusoloSmart reduce our Contact Centre queue times during peak filing deadlines?

Defensible Response:

“Over 70% of inbound inquiries during deadline weeks are repetitive procedural questions: checking filing due dates, verifying the PAYE threshold, requesting TIN recovery steps, or inquiring about bank payment registration numbers (PRNs). OmusoloSmart resolves these Level-1 inquiries instantly with zero human intervention in sub-second time ($p_{50} < 700\text{ms}$). This deflects thousands of calls from the queue, allowing human officers to focus on high-value taxpayer audits, debt negotiations, and complex exemptions.”

Q3: What happens when a taxpayer is angry, complaining about delayed refunds or unexpected tax penalties?

Defensible Response:

“The platform includes real-time emotional intelligence (EQ) scoring and distress detection. When a message contains sentiment cues indicating financial hardship, anger, or penalty anxiety, the bot suppresses cheerful marketing tones. It opens with an empathetic acknowledgment (*‘I understand this is stressful — let’s look at your options’*), provides the statutory appeal or payment plan procedures under the Tax Procedures Code Act, and generates an automated warm escalation ticket containing the entire diagnostic context for a URA Contact Centre officer.”

Q4: Can the taxpayer switch languages midway through a conversation?

Defensible Response:

“Yes. OmusoloSmart supports dynamic multi-turn language switching. A taxpayer can begin an inquiry in English, transition to Luganda for procedural clarification, and conclude in Swahili. The context manager preserves the rolling conversation state and entities across language boundaries without losing context or restarting the dialogue.”

Q5: Does this replace our existing URA Contact Centre staff?

Defensible Response:

“No. OmusoloSmart is designed as a **force multiplier**, not a replacement. It handles routine Level-1 informational triage 24/7. When a taxpayer faces enforcement notices, audits, or disputes, the bot packages the context and creates a warm handoff to URA officers, ensuring human agents focus their expertise where empathy, negotiation, and discretion are required. Furthermore, Contact Centre officers can use OmusoloSmart as an internal co-pilot to instantly retrieve verified citations during live calls.”

9.2 Category B: Statutory Accuracy, Legal Reliability & Liability

Q6: How can we be certain the AI won’t invent a tax rate or give unauthorized tax advice?

Defensible Response:

“OmusoloSmart enforces **three layers of epistemic defense**:

1. *Deterministic Calculator Router*: Statutory rate lookups (VAT, PAYE, WHT, Corporation Tax) bypass generative guessing entirely and pull directly from the versioned FY2026/27 URA rate table.
2. *Entailment & Contradiction Withholding*: If an LLM-drafted response contradicts cited URA documentation (e.g., drafting 20% when the statute prescribes 18%), the system strictly withholds the response, serves the official passage, and alerts an officer.
3. *Zero Generative Drift*: The model operates at greedy zero-temperature with statutory prompt guardrails preventing speculative legal interpretation.

”

Q7: If the Minister of Finance amends a tax rate in the National Budget in June, how do we update the assistant, and how long does it take?

Defensible Response:

“Unlike deep learning models that require months of retraining or fine-tuning, OmusoloSmart decouples **knowledge** from **reasoning**. Statutory rates live in declarative fiscal-year rate tables (`tables.py`). Updating an excise duty rate or changing a threshold is a single-line configuration change verified by automated test suites in seconds. Superseded handbooks are automatically discounted in retrieval ranking.”

Q8: How does the assistant protect URA from legal liability if a taxpayer relies on its answers?

Defensible Response:

“OmusoloSmart enforces a strict epistemic disclaimer framework. Every substantive response cites its statutory basis (e.g., Section 5 of the Value Added Tax Act, Section 83 of the Income Tax Act). If the claim verification engine detects that a response has low grounding

confidence (< 0.65) or an uncited factual assertion, it suppresses the generated text and serves a pre-verified deterministic passage directly from the official URA handbook, accompanied by official URA Contact Centre verification channels.”

9.3 Category C: Language Models, Vernacular Dialects & Voice AI

Q9: Why did you choose Sunbird’s Sunflower and SALT models over commercial alternatives like OpenAI’s GPT-4 or Google Gemini?

Defensible Response:

“Two critical reasons: **linguistic sovereignty** and **data privacy**. Commercial American models struggle with Ugandan indigenous languages, frequently inventing Swahili-shaped pseudo-words when asked for Luganda, and they require sending sensitive Ugandan taxpayer data outside the country’s borders. Sunbird AI’s Sunflower-14B, Whisper-SALT, and Spark-TTS-SALT were trained locally on Ugandan speech and text corpora by Ugandan researchers at Makerere, ensuring authentic dialect comprehension, correct pronunciation of local names and institutions, and strict compliance with national data sovereignty.”

Q10: How does it perform with real Ugandan dialects, street tax slang, or code-switching between English and Luganda?

Defensible Response:

“The language stack uses **Sunbird AI’s SALT models**, trained on verified Ugandan speech and text corpora from Makerere University. It natively tolerates code-switching (*‘Luglish’*), vernacular noun-class morphology, and colloquial financial expressions (*‘obukadde 5’*, *‘siringi emitwalo ebiri’*), without defaulting to European English grammatical conventions.”

Q11: Does the voice system understand people speaking in noisy environments, like a busy market in Kikuubo or near a taxi park?

Defensible Response:

“Whisper-SALT was fine-tuned specifically on local Ugandan acoustic datasets, community radio broadcasts, and field speech recordings gathered across Uganda by Makerere and Sunbird AI. It exhibits high acoustic robustness against background market noise, vehicle traffic, and colloquial code-switching compared to standard international speech engines, operating with an average Real-Time Factor of 0.129x.”

9.4 Category D: IT Infrastructure, System Integration & PDPO Compliance

Q12: Where does taxpayer data live, and how do we comply with the Data Protection and Privacy Act 2019 (PDPO)?

Defensible Response:

“Yes. OmusoloSmart incorporates privacy-by-design:

- **Automated PII Redaction:** National Identification Numbers (NINs), phone numbers, bank details, and passwords are sanitized at the input boundary before hitting inference or logging.
- **Data Sovereignty:** The entire software stack runs on local Ugandan hardware (Tier-III data centers or Crane Cloud). No data is exported abroad.
- **Cryptographic Audit Ledger:** Every query, response, and officer handoff is recorded in an immutable Merkle tree ledger, ensuring tamper-evident accountability.

”

Q13: How easily does this integrate into our existing support channels (WhatsApp, Web Portal, IVR)?

Defensible Response:

“The backend provides standardized REST endpoints, W3C Server-Sent Events (SSE), and duplex WebSockets (`/v1/chat`, `/v1/chat/stream`, `/v1/voice/chat/stream`). It integrates into the URA web portal as a lightweight React/Next.js widget,

connects to WhatsApp Business API via standard webhooks, and connects to telephony switches (SIP / Asterisk / FreeSWITCH) via standard audio streaming formats (PCM/WAV/MP3).”

Q14: Can the system function in remote border posts or regional offices with unstable or absent internet connectivity?

Defensible Response:

“Yes. In addition to the cloud/gateway microservices stack, OmusoloSmart includes an exportable, standalone offline engine. Powered by an embedded FAISS vector index and local SQLite database, the offline engine runs on edge laptops or desktop workstations in remote border posts (such as Malaba, Busia, or Mutukula) without requiring active internet connectivity.”

10 Presenter Checklist & Engagement Protocol

10.1 Technical Pre-Flight Verification (T-60 Minutes)

- ✓ **Public Gateway Liveness:** Execute health check to confirm HTTP 200 response:
`curl -s https://struttingly-nongeological-briella.ngrok-free.dev/api/health`
- ✓ **GPU Stack Readiness:** Verify that vLLM (Sunflower-14B-FP8), Whisper-SALT, and Spark-TTS-SALT are active with VRAM allocated on the designated GPU host.
- ✓ **Audio Peripherals:** Test microphone input levels and ensure an external speaker is connected for the audience to clearly hear Luganda and Swahili voice synthesis.
- ✓ **Cache Warmup:** Run a test query in each language to ensure inference paths and speech models are warm.
- ✓ **Artifacts on Hand:** Keep evaluation benchmark reports and logs readily available to project empirical proof points if requested.

10.2 Audience Engagement Strategy

- **Emphasize Collaboration:** Frame OmusoloSmart not as an external critique of current operations, but as an advanced digital tool engineered specifically to empower URA Client Experience officers.
- **Focus on FCR and Informal Sector:** Constantly tie technological features (like multilingual speech and deterministic grounding) back to URA’s core metrics: First Contact Resolution, queue reduction, and broadening the national tax base.
- **Invite Live Spontaneous Queries:** Encourage URA officers in the room to test their hardest statutory edge cases live during the open floor session.

10.3 Proposed Next Steps & Engagement Roadmap

1. **Phase 1 Pilot Deployment:** Integrate OmusoloSmart as a virtual support pilot on `ura.go.ug` and URA’s official WhatsApp channel for a 60-day trial covering Domestic Taxes and Instant TIN support.
2. **Contact Centre Agent Assist Mode:** Deploy OmusoloSmart to Contact Centre agents as an internal co-pilot to instantly retrieve verified citations and calculate complex assessments during live calls.
3. **DRMS Regional Outreach Expansion:** Package the offline multimodal voice engine for deployment at mobile tax clinics, border posts, and regional tax education hubs across Uganda.

Report Artifact References: `docs/Reports/MULTILINGUAL_2000_FAQS_EVALUATION_REPORT.md` • `docs/presentation/multilang_concurrency`

Benchmarks: 99.85% Accuracy (1,997/2,000 FAQs) • Multilingual Concurrency Audit (100% Availability across Load, Volume, Stress, Spike) • School of Computing and Informatics Technology, Makerere University.